

# MTE 203 – Advanced Calculus

## Week 2 Tutorial

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### Problem 1 [S. 11.5, P.7]

Find the equation of the plane containing the line,

$$x - y + 2z = 4$$

$$2x + y + 3z = 6$$

and the point  $(1, -2, 4)$ .

### Problem 2 [S. 11.6, P.27]

Find the distance between the skewed lines<sup>1</sup> defined by the intersection of the planes,

$$x + y - z = 4$$

$$2x - z = 4$$

And the line,

$$x = \frac{y+1}{2} = \frac{z-1}{3}$$

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<sup>1</sup> The distance between skew lines was solved in Lecture 3 and it is also solved in Example 24 of your textbook.